CHAPTER 5

Function Operations

5

SYLLABUS 2.5 2.10 2.11 2.14–2.16

Hold a card in front of a lamp and look at its shadow on the wall. Slide the card and the shadow slides. Move it closer to the lamp and the shadow grows. Turn it over and the shadow flips. The card itself never changes. Only what you do to it changes.

Graphs work the same way, and that is what makes this part of the course manageable. There are only a few shapes worth learning by heart, a line, a parabola, a square root, a reciprocal, together with a short list of moves that slide them, stretch them and reflect them. Every other curve you will meet is one of those few shapes with some of those moves applied to it.

So this chapter has two halves. The first is the list of moves, and what each one does to an equation. The second is **composition**: the rule for feeding the output of one function straight into the next, which is how those moves are chained together in the first place.

By the end of this chapter you will be able to compose two functions and find the domain of the result, and to describe and apply translations, stretches and reflections. You will solve equations by reading intersections off a graph, test a function for symmetry, and sketch the related graphs $|f(x)|$, $f(|x|)$ and $\dfrac{1}{f(x)}$. What ties these together is that every number in a formula corresponds to something you can see. Change the number and the graph moves in a way you can predict. A formula and a picture are two views of one function, and this chapter is about reading each from the other.

5.1 Composite Functions

Two functions can be joined end to end. Send a number into the first function. Take the number that comes out. Send that number into the second function. The single rule that carries out both steps at once is called a **composite** function.

$$x \longrightarrow \boxed{g} \longrightarrow g(x) \longrightarrow \boxed{f} \longrightarrow f(g(x))$$

Figure 5.1 Composition sends x through g first, then feeds the result into f. The inner function acts first, which is why $f \circ g$ is read from the inside out.

> DEF The **composite** $f \circ g$ is the function $(f \circ g)(x) = f(g(x))$: apply g first, then f. The **identity function** $I(x) = x$ returns every input unchanged.

The symbol $\circ$ is read "of", so $f \circ g$ is "f of g". It is also read "f after g", which says the same thing in the order the functions act. It is not multiplication. Read the notation from right to left: the function written closest to x is the one that acts on x first.

The key to composition is that the x in $f(x)$ is only a **placeholder**. It marks the position where the input goes. It does not have to be a single letter, and it does not have to be a number.

Read f as a rule with an empty slot. If $f(x) = x^2 + 1$, the rule says *square the input, then add one*. So $f(3) = 3^2 + 1$, and $f(2x) = (2x)^2 + 1$, and $f(\text{anything}) = (\text{anything})^2 + 1$. The input can be an expression, and it can be another function:

$$f(g(x)) = \big(g(x)\big)^2 + 1.$$

Nothing new is happening here. You are substituting a whole function exactly as you would substitute a number.

To build $(f \circ g)(x)$, apply g to x, then replace every x in the formula for $f(x)$ by the whole expression $g(x)$, then simplify.

The order changes the answer. Putting on socks then shoes is not the same as shoes then socks, and composing two functions behaves the same way.

EXAMPLE 5.1

For $f(x) = x^2 + 1$ and $g(x) = 2x$, find $(f \circ g)(x)$ and $(g \circ f)(x)$.

For $f \circ g$, apply g first. Its output is $2x$, so substitute $2x$ for every x in $f(x) = x^2 + 1$, then simplify:

$$(f \circ g)(x) = f(2x) = (2x)^2 + 1 = 4x^2 + 1.$$

For $g \circ f$, apply f first. Its output is x^2+1, so substitute x^2+1 for every x in $g(x) = 2x$, then simplify:

$$(g \circ f)(x) = g(x^2+1) = 2(x^2+1) = 2x^2+2.$$

The two answers are different. In general $f \circ g \neq g \circ f$, so always check which function acts first.

Composition also explains what an inverse does. The **inverse** of f, written f^{-1}, is the function that reverses f: if f sends 3 to 10, then f^{-1} sends 10 back to 3. Apply f and then apply f^{-1}, and you arrive back at the number you started with. The composite is therefore the identity function.

KEY For a function f with inverse f^{-1},

$$(f^{-1} \circ f)(x) = x \qquad \text{and} \qquad (f \circ f^{-1})(x) = x.$$

An inverse exists exactly when f is **one-to-one**: no two inputs share an output.

YOUR TURN 5.1 Composite Functions

1. For $f(x) = x+3$ and $g(x) = 2x$, find each composite.

(a) $(f \circ g)(x)$ [2] (b) $(g \circ f)(x)$ [2]
(c) $(f \circ f)(x)$ [2] (d) $(g \circ g)(x)$ [2]

2. For $f(x) = x^2$ and $g(x) = x-1$, find each of the following.

(a) $(f \circ g)(x)$ [2] (b) $(g \circ f)(x)$ [2]
(c) $(f \circ g)(3)$ [1] (d) $(g \circ f)(3)$ [1]

3. For $f(x) = x+5$ and $g(x) = x^2$, evaluate each of the following.

(a) $(g \circ f)(2)$ [2] (b) $(f \circ g)(2)$ [2]
(c) $(g \circ f)(-3)$ [2] (d) $(f \circ g)(-3)$ [2]

4. For $f(x) = \dfrac{1}{x}$ and $g(x) = x+1$, find each composite and state its domain.

(a) $(f \circ g)(x)$ [2] (b) the domain of $f \circ g$ [2]
(c) $(g \circ f)(x)$ [2] (d) the domain of $g \circ f$ [2]

5. Find the inverse of each function.

(a) $f(x) = 3x-2$ [2] (b) $f(x) = \dfrac{x}{4}+1$ [3]

(c) $f(x) = \dfrac{x-5}{2}$ [2]

(d) $f(x) = \dfrac{1}{x-1}$ [3]

6. Let $f(x) = 2x + 1$.

(a) Find $f^{-1}(x)$. [2]
(b) Show that $(f \circ f^{-1})(x) = x$. [2]
(c) Show that $(f^{-1} \circ f)(x) = x$. [2]
(d) State what these two results tell you about f^{-1}. [1]

7. Let $f(x) = x^2$ and $g(x) = x + 2$.

(a) Find $(f \circ g)(x)$ and $(g \circ f)(x)$, and show that they are not the same. [3]
(b) Find the value of x for which the two composites are equal. [3]

5.2 Transformations of Graphs

A graph is an object you can move. There are three kinds of move. A **translation** slides it, a **reflection** flips it, and a **stretch** scales it.

Each move matches one small change in the equation. Learn which change produces which move, and you can read any transformed equation directly.

Move	Equation	Acts on	Effect on the graph
Translation	$y = f(x) + k$	output	up by k, or down if $k < 0$
Translation	$y = f(x - h)$	input	right by h, or left if $h < 0$
Reflection	$y = -f(x)$	output	in the x-axis
Reflection	$y = f(-x)$	input	in the y-axis
Stretch	$y = a\,f(x)$	output	vertically, factor a
Stretch	$y = f(bx)$	input	horizontally, factor $\frac{1}{b}$

Read the third column and the pattern appears. Everything depends on whether the change sits *outside* the bracket or *inside* it: a change to the output is always vertical, and a change to the input is always horizontal.

A change outside the bracket acts on the output, after f has been applied. It behaves as you would expect. Adding k raises every value by k, so the graph moves up by k. Multiplying by a multiplies every value by a, so the graph stretches vertically by a.

A change inside the bracket acts on the input, before f is applied, so its effect is reversed. Take $y = f(x-h)$. At $x = p+h$ this returns $f(p)$, which is the value the original graph had at $x = p$. Every point therefore moves h units to the *right*, not left. In the same way, $y = f(bx)$ returns $f(p)$ at $x = \frac{p}{b}$, so every point moves from p to $\frac{p}{b}$. That is a horizontal stretch of factor $\frac{1}{b}$, which compresses the graph towards the y-axis whenever $b > 1$.

Translations

A translation slides the whole graph. Its shape and its size do not change, only its position.

Vertically, $y = f(x) + k$ adds k to every output, so every point moves up by k. A negative k moves it down.

Horizontally, $y = f(x-h)$ moves the graph right by h. This is the sign students most often reverse, so it is worth holding on to the reason above: the change is inside the bracket, so it acts on the input, and its effect is the opposite of its sign.

In examination questions a translation is usually written as a vector. The graph of $y = f(x-h) + k$ is the graph of $y = f(x)$ translated by $\binom{h}{k}$.

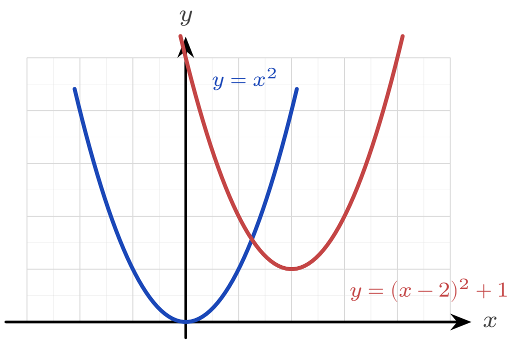

Figure 5.2 The graph of $y = (x-2)^2 + 1$ is the graph of $y = x^2$ translated 2 units right and 1 unit up. The vertex moves from $(0,0)$ to $(2,1)$, and the shape is unchanged.

EXAMPLE 5.2

Describe how the graph of $y = (x-2)^2 + 1$ is obtained from $y = x^2$.

Start with the bracket. The -2 sits inside it, so it acts on the input and moves the parabola 2 units *right*.

Now take the rest. The $+1$ sits outside the bracket, so it acts on the output and moves the parabola 1 unit *up*.

Together these send the vertex from $(0,0)$ to $(2,1)$, which is exactly what the completed-square form predicts. Transformations and the completed-square form are two descriptions of the same thing.

Reflections

A reflection flips the graph across an axis. Every point keeps its distance from that axis and moves to the other side of it.

For $y = -f(x)$ the sign of every output changes, so the point (p, q) moves to $(p, -q)$. That is a reflection in the x-axis. A maximum becomes a minimum, and any point already on the x-axis stays where it is.

For $y = f(-x)$ the sign of every input changes, so the point (p, q) moves to $(-p, q)$. That is a reflection in the y-axis. The y-intercept does not move, because changing the sign of 0 leaves it as 0.

EXAMPLE 5.3

Describe how the graph of $y = -(x-1)^2$ is obtained from $y = x^2$.

There are two changes. The -1 is inside the bracket, so it is horizontal: translate 1 unit right. The minus sign is outside the bracket, so it is vertical: reflect in the x-axis.

One change is horizontal and the other is vertical, so they do not affect each other and may be done in either order.

The vertex moves from $(0, 0)$ to $(1, 0)$, and the parabola now opens downward.

Stretches

A stretch changes the size of the graph in one direction only. One line stays fixed while everything else moves away from it or towards it.

For $y = a\,f(x)$ every output is multiplied by a, so (p, q) moves to (p, aq). This is a vertical stretch of factor a. Points on the x-axis do not move, because $a \times 0 = 0$.

For $y = f(bx)$ every input is divided by b, as shown above, so (p, q) moves to $(\frac{p}{b}, q)$. This is a horizontal stretch of factor $\frac{1}{b}$. Points on the y-axis do not move.

CAUTION *A factor smaller than 1 is still called a stretch. The graph of $y = \frac{1}{2}f(x)$ is a vertical stretch of factor $\frac{1}{2}$, even though it makes the graph shorter. Write "stretch, factor $\frac{1}{2}$", not "squash".*

EXAMPLE 5.4

The graph of $y = f(x)$ has a maximum point at $(6, 4)$. State the coordinates of that maximum point on the graph of (a) $y = 3f(x)$, (b) $y = f(3x)$.

(a) The 3 is outside the bracket, so it is a vertical stretch of factor 3. It multiplies the y-coordinate and leaves the x-coordinate alone: $(6, 12)$.

(b) The 3 is inside the bracket, so it is a horizontal stretch of factor $\frac{1}{3}$. It divides the x-coordinate and leaves the y-coordinate alone: $(2, 4)$.

Each stretch changes one coordinate and leaves the other unchanged. Checking which coordinate moved is a quick way to confirm you have used the right one.

Composite transformations

Most graphs are several moves away from their parent function, applied one after another. This is a **composite transformation**. Describing one is a matter of taking the changes in a reliable order.

KEY To describe a composite transformation:

1. Split the changes into two groups: those **inside** the bracket, which are horizontal, and those **outside** it, which are vertical.
2. Do the two groups in either order. A horizontal move and a vertical move do not affect each other.
3. Inside each group the order matters. Apply the **stretch before the translation**.

Step 3 is the one to understand rather than memorise. A stretch multiplies everything it is given, including a translation that has already been carried out. Do the stretch first and it has only the original graph to multiply. The second example below shows exactly what goes wrong in the other order.

It also helps to write the target equation in one standard form before you start:

$$y = a\,f(b(x-h)) + k.$$

Then a is the vertical stretch factor, b gives the horizontal stretch factor $\frac{1}{b}$, h is the horizontal translation and k is the vertical translation, and every value can be read directly.

CAUTION *Factorise inside the bracket first. In $y = f(2x-6)$ the translation is not 6. Written as $y = f(2(x-3))$, it is a horizontal stretch of factor $\frac{1}{2}$ followed by a translation of 3 to the right. Forgetting to factorise is the commonest error in this topic.*

EXAMPLE 5.5

Describe a sequence of transformations that maps $y = \sqrt{x}$ onto $y = 2\sqrt{x+3} - 1$.

Take the change inside the bracket first. The $+3$ acts on the input, so it translates the graph 3 units *left* and gives $\sqrt{x+3}$.

Now take the vertical changes, stretch before translation. The factor 2 stretches the graph vertically by scale factor 2 and gives $2\sqrt{x+3}$. The -1 then translates it 1 unit *down*.

So the sequence is: translate 3 left, stretch vertically by factor 2, translate 1 down.

Check it by tracking the endpoint. The graph of $y = \sqrt{x}$ begins at $(0,0)$. The translation carries it to $(-3,0)$. The stretch leaves it there, because $2 \times 0 = 0$. The final translation carries it to $(-3,-1)$, which is exactly where $y = 2\sqrt{x+3} - 1$ begins.

The same care is needed inside the horizontal group, where a stretch and a translation also fail to commute. Within either group the order really does matter. Some pairs of moves give the same result in either order, and such a pair is said to **commute**. Other pairs do not, and swapping those produces a different curve.

EXAMPLE 5.6

Starting from $y = x^2$, apply a vertical stretch by factor 2 and a translation 1 unit up, in both orders. Are the results the same?

Stretch first, then translate. The stretch doubles every value, giving $2x^2$; the translation then adds

1:

$$x^2 \longrightarrow 2x^2 \longrightarrow 2x^2 + 1.$$

Translate first, then stretch. The translation adds 1, giving x^2+1; the stretch then doubles everything, *including* the 1:

$$x^2 \longrightarrow x^2 + 1 \longrightarrow 2(x^2 + 1) = 2x^2 + 2.$$

The two curves are different. A vertical stretch multiplies whatever is already there, so a translation applied first is multiplied as well. A vertical stretch and a vertical translation commute only when one of them does nothing, that is when the stretch factor is 1 or the translation is 0. In every other case you must state them in the order that actually builds the target curve.

TIP *In examinations, "describe the sequence of transformations" expects named moves with their values: translation by $\binom{-3}{0}$, vertical stretch scale factor 2, translation by $\binom{0}{-1}$. Vague words like "moved" or "squashed" lose the marks.*

YOUR TURN 5.2 Transformations of Graphs

8. Describe the single transformation that takes $y = x^2$ to each graph.

(a) $y = x^2 + 3$ [1] (b) $y = (x + 4)^2$ [1]
(c) $y = -x^2$ [1] (d) $y = (x - 1)^2$ [1]

9. Describe the single transformation that takes $y = f(x)$ to each graph.

(a) $y = f(-x)$ [1] (b) $y = -f(x)$ [1]
(c) $y = 3f(x)$ [1] (d) $y = f(2x)$ [2]

10. The point $(2, 5)$ lies on $y = f(x)$. Find the coordinates of its image on each graph.

(a) $y = f(x - 3) + 1$ [2] (b) $y = 3f(x)$ [1]
(c) $y = f(2x)$ [2] (d) $y = -f(x) + 2$ [2]

11. The point $(-2, 3)$ lies on $y = f(x)$. Find the coordinates of its image on each graph.

(a) $y = f(2x)$ [2] (b) $y = f(x) - 4$ [1]
(c) $y = f(-x)$ [1] (d) $y = \frac{1}{2}f(x)$ [2]

12. Describe the sequence of transformations that maps $y = \sqrt{x}$ onto each graph.

(a) $y = \sqrt{x + 3}$ [2] (b) $y = 2\sqrt{x}$ [2]

(c) $y = 2\sqrt{x+3} - 1$ [3]

(d) $y = \sqrt{2x-6}$ *(Hint: factorise inside the root first)* [4]

13. Start from $y = x^2$.

(a) Apply a vertical stretch of factor 3, then a translation 2 units up. Write the resulting equation. [2]

(b) Apply the same two moves in the opposite order, and write the resulting equation. [2]

14. Consider $y = f(4x-8)$.

(a) Write it in the form $y = f(b(x-h))$. [2]

(b) Describe the sequence of transformations from $y = f(x)$. [3]

5.3 Solving Equations Graphically

Every equation can be read as a question about two graphs. Solving $f(x) = g(x)$ means finding the values of x at which f and g return the same output. Those are exactly the points where the two graphs cross, because a crossing point lies on both curves at once.

So: sketch both graphs, find the crossings, and read their x-coordinates. Those x-coordinates are the solutions.

This matters most when no algebraic method exists. The equation $e^x = 3 - x$ has no solution in **closed form** (written as an exact formula using standard functions). But $y = e^x$ and $y = 3 - x$ clearly cross exactly once, and a calculator locates that crossing to as many decimal places as you need. Here **technology** is not a way of avoiding the mathematics. It is the method.

EXAMPLE 5.7

Solve $x^2 = x + 2$, and then solve $x^2 \leq x + 2$.

Start with the equation. Collect everything on one side so it can be factorised, then solve:

$$x^2 - x - 2 = 0 \implies (x-2)(x+1) = 0 \implies x = -1 \text{ or } x = 2.$$

These two values are where the parabola $y = x^2$ meets the line $y = x + 2$.

Now the inequality. It asks for the values of x where the parabola is *below* the line, or level with it. The crossings at $x = -1$ and $x = 2$ divide the x-axis into three parts, and the parabola is below the line only in the middle part:

$$x^2 \leq x + 2 \quad \text{for} \quad -1 \leq x \leq 2.$$

This is the pattern for every inequality of this kind. Solving the equation finds the boundary points. The graph then shows which side of them to keep.

EXAMPLE 5.8

After a Finnish sauna is switched off, its temperature in °C after t minutes is modelled by $T(t) = 20 + 70e^{-0.12t}$. Find how long it takes the sauna to cool to $40\,^\circ$C.

Read the question as two graphs: the cooling curve $y = T(t)$, and the horizontal line $y = 40$. The answer is the t-coordinate where they cross. Graph both on a graphic display calculator (GDC) and use intersect, which gives $t \approx 10.4$.

The algebra confirms it. Substitute $T(t) = 40$ and solve for t:

$$70e^{-0.12t} = 20 \implies e^{-0.12t} = \tfrac{2}{7} \implies t = \frac{\ln(7/2)}{0.12} \approx 10.4 \text{ minutes.}$$

The model also tells you something without any solving at all. As t grows, $e^{-0.12t}$ approaches 0, so $T(t)$ approaches 20. That 20 is the room temperature: the line $y = 20$ is a horizontal asymptote, and the sauna cools towards it without ever reaching it.

Inequalities between functions HL

The solution of $g(x) \geq f(x)$ is the set of x values for which the graph of g lies on or above the graph of f.

Work in two stages. First find the **boundary points**: the values of x where the two graphs meet, and also the values where either function is undefined. Then test each interval between them to see whether the inequality holds there.

Both kinds of boundary point matter. A vertical asymptote separates one interval from the next just as effectively as a crossing does, even though the graphs never meet there.

EXAMPLE 5.9

Solve $\dfrac{x+2}{x-1} \geq 3$.

Do not multiply both sides by $x-1$. Its sign depends on x: it is negative when $x < 1$ and positive when $x > 1$. Multiplying an inequality by a negative number reverses the sign, so this method splits into cases and is easy to get wrong.

Instead, collect everything on one side and combine it into a single fraction:

$$\frac{x+2}{x-1} - 3 \geq 0 \implies \frac{(x+2) - 3(x-1)}{x-1} \geq 0 \implies \frac{5-2x}{x-1} \geq 0.$$

A fraction changes sign only where it is zero or where it is undefined, so these are the boundary points. It is zero when $5 - 2x = 0$, that is $x = \frac{5}{2}$, and undefined when $x - 1 = 0$, that is $x = 1$.

These two values split the x-axis into three intervals. Test the sign of $\dfrac{5-2x}{x-1}$ on each:

$$x < 1: \ \tfrac{+}{-} < 0, \qquad 1 < x < \tfrac{5}{2}: \ \tfrac{+}{+} > 0, \qquad x > \tfrac{5}{2}: \ \tfrac{-}{+} < 0.$$

Only the middle interval satisfies the inequality, so the solution is

$$1 < x \le \tfrac{5}{2}.$$

Check the two endpoints separately. Exclude $x = 1$, because the fraction is undefined there. Include $x = \frac{5}{2}$, because the inequality allows equality and the fraction is zero there.
A GDC graph of $y = \frac{x+2}{x-1}$ against $y = 3$ confirms it: the curve lies above the line only between the asymptote and the crossing.

CAUTION *Never multiply an inequality by an expression whose sign you do not know. Move everything to one side, combine into a single fraction, and read the sign from the critical values.*

YOUR TURN 5.3 Solving Equations Graphically

15. Solve each equation.

(a) $x^2 = 2x + 3$ [2] (b) $x^2 = 4 - 3x$ [2]
(c) $x^2 = x + 6$ [2] (d) $x^2 = 9$ [1]

16. Solve each inequality.

(a) $x^2 < 4$ [2] (b) $x^2 \ge 9$ [2]
(c) $x^2 \le x + 6$ [3] (d) $x^2 > 2x + 3$ [3]

17. Find the coordinates of every intersection point of each pair of graphs.

(a) $y = x^2$ and $y = x + 6$ [3] (b) $y = x^2$ and $y = 2x + 8$ [3]
(c) $y = x^2 - 1$ and the x-axis [2] (d) $y = x^2$ and $y = -x$ [3]

18. Consider the equation $e^x = 3 - x$.

(a) Explain why it has exactly one real solution. [2]
(b) Find that solution, correct to three significant figures. [2]
(c) State why a graph is the appropriate method for this equation. [2]

19. Solve each inequality. HL

(a) $\dfrac{x+2}{x-1} \ge 3$ [4] (b) $\dfrac{2x-1}{x+2} \le 1$ [4]

20. After a sauna is switched off, its temperature in °C after t minutes is $T(t) = 20 + 70e^{-0.12t}$.

(a) State the temperature at the moment it is switched off. [1]

(b) Find the temperature after 5 minutes. [2]

(c) Find how long it takes to cool to $40\,^{\circ}$C. [3]

(d) State the temperature the sauna approaches, and explain what it represents. [2]

5.4 Symmetry, Modulus and Related Graphs HL

Some graphs repeat themselves. If you know one half of such a graph, the symmetry determines the other half, so you only need to sketch half of it. It also reveals structure that the formula alone does not show.

Type	Test	Symmetry of the graph	Examples
Even	$f(-x) = f(x)$	in the y-axis	x^2, x^4, $\lvert x\rvert$, $\cos x$
Odd	$f(-x) = -f(x)$	rotational, order two, about the origin	x, x^3, $\frac{1}{x}$, $\sin x$
Neither	neither test holds	no symmetry of either kind	$x^2 + x$, $x + 1$, e^x

The names come from the powers. The functions $x^2, x^4, \ldots$ have even exponents and satisfy $f(-x) = f(x)$; the functions $x, x^3, \ldots$ have odd exponents and satisfy $f(-x) = -f(x)$. Most functions are neither, so "neither" is a complete and correct answer when the tests fail.

To test a function, substitute $-x$ for x, simplify, and compare the result with the original.

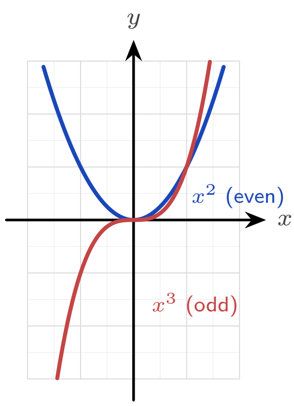

Figure 5.3 $y = x^2$ is even: reflecting it in the y-axis leaves it unchanged. $y = x^3$ is odd: reflecting it in the y-axis and then in the x-axis leaves it unchanged, which is a rotation of 180° about the origin.

A third kind of symmetry belongs to functions that are their own inverse. Recall that f^{-1} reverses f. If applying f twice already returns the input, so that $f(f(x)) = x$, then f reverses itself and $f = f^{-1}$. Such a function is called **self-inverse**, and its graph is symmetric in the line $y = x$.

EXAMPLE 5.10

Show that $f(x) = \dfrac{1}{x}$ is self-inverse, and sketch its graph.

Apply the function twice. Substitute the whole expression $\frac{1}{x}$ in place of x:

$$f(f(x)) = \frac{1}{\frac{1}{x}} = x.$$

Applying f twice returns the input, so f reverses itself and $f = f^{-1}$.

The sketch shows what this means geometrically. Reflecting any graph in the line $y = x$ produces the graph of its inverse. Since f *is* its own inverse, the reflection must leave the curve unchanged, so the curve is symmetric in $y = x$.

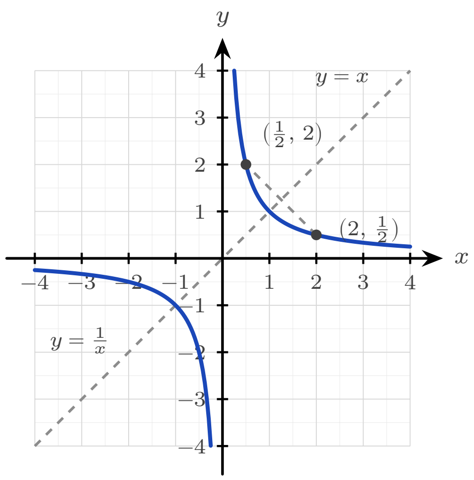

Figure 5.4 The reciprocal is symmetric in the line $y = x$. Each point on the curve has a partner with its coordinates swapped, so reflecting in $y = x$ returns the same curve.

Self-inverse functions are not rare, and they are not all this simple. The reflection $f(x) = 6 - x$ is self-inverse, and so is $f(x) = \dfrac{2x+1}{x-2}$, which you can check by the same substitution.

EXAMPLE 5.11

Determine whether $f(x) = x^3 - x$ is odd, even, or neither.

Substitute $-x$ for x, simplify, then look for a factor of -1:

$$f(-x) = (-x)^3 - (-x) = -x^3 + x = -(x^3 - x) = -f(x).$$

The result is $-f(x)$, so by the definition the function is **odd**. Its graph therefore has rotational symmetry of order two about the origin.

CAUTION *A function like $f(x) = x^2$ is not one-to-one on all of $\mathbb{R}$, since $f(2) = f(-2)$, so it has no inverse there. Restricting the domain to $x \geq 0$ makes it one-to-one, and the inverse is $f^{-1}(x) = \sqrt{x}$.*

Restricting the domain to make an inverse exist is a standard examination technique. Note what the caution above shows: the restriction you choose decides which branch of the inverse you get.

EXAMPLE 5.12

Let $f(x) = (x-3)^2$ for $x \geq 3$. Find $f^{-1}(x)$, stating its domain.

First check that an inverse exists. On $x \geq 3$ the parabola only rises, so no two inputs share an output and f is one-to-one.

Now find the inverse. Write $y = f(x)$ and solve for x:

$$x - 3 = \pm\sqrt{y} \implies x = 3 \pm \sqrt{y}.$$

Two branches appear, so the restriction must decide between them. Since $x \geq 3$, the expression $x - 3$ cannot be negative, so only the $+$ branch is possible. Hence

$$f^{-1}(x) = 3 + \sqrt{x}, \qquad x \geq 0,$$

where the domain of f^{-1} is the range of f. Had the restriction been $x \leq 3$ instead, the same algebra would give $f^{-1}(x) = 3 - \sqrt{x}$. The algebra is the same until that final choice of sign, but the inverse is different.

One last kind of repetition. A function is **periodic** if its graph repeats at regular intervals, that is, if $f(x+p) = f(x)$ for some fixed positive number p. The smallest such p is called the **period**. The trigonometric functions are the most important example: their graphs repeat forever along the x-axis.

The modulus $|f(x)|$

The remaining graphs in this section are also produced by symmetry. Each is built from a function f by one operation, and each has a single visual rule, so you are never expected to plot points.

The modulus of a number is its size without its sign, so $|f(x)|$ is never negative.

That gives the rule at once. Any part of the graph of f already above the x-axis is unchanged. Any part below the axis has its sign flipped, so it is reflected upward in the x-axis. The x-intercepts do not move, because $|0| = 0$.

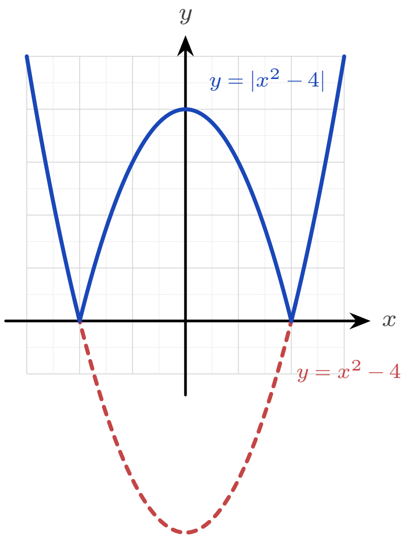

Figure 5.5 $y = |x^2 - 4|$ against $y = x^2 - 4$. The part of the parabola below the x-axis is reflected upward; everything already above the axis is untouched, so the two roots become corners.

Related graphs

Three more operations appear on the syllabus, and each follows one idea.

For $y = f(|x|)$ the modulus is applied to the *input*, so f receives the same number whether x is positive or negative. Keep the right-hand half of the graph and reflect it in the y-axis to make the left-hand half. The result is always an even function.

For $y = [f(x)]^2$ every output is squared, so no output is negative. Each root of f becomes a point where the new curve touches the x-axis from above rather than crossing it.

For $y = f(ax + b)$ the change is inside the bracket, so it is a combination of the horizontal transformations of §5.2.

YOUR TURN 5.4 Symmetry, Modulus and Related Graphs HL

21. Determine whether each function is odd, even, or neither.

(a) $f(x) = x^4 - 3x^2$ [2]
(b) $f(x) = x^3 + x$ [2]
(c) $f(x) = x^2 + x$ [1]
(d) $f(x) = x^5$ [1]
(e) $f(x) = 5$ [1]
(f) $f(x) = \dfrac{1}{x}$ [2]

22. Self-inverse functions.

(a) Show that $f(x) = \dfrac{1}{x}$ is self-inverse. [2]
(b) Show that $f(x) = 6 - x$ is self-inverse. [2]

(c) Show that $f(x) = \dfrac{2x+1}{x-2}$ is self-inverse. [3]

(d) Explain why $f(x) = 2x$ is not self-inverse. [2]

23. Restricting a domain so that an inverse exists.

(a) State a domain on which $f(x) = x^2$ has an inverse, and give that inverse. [2]

(b) Do the same for $f(x) = (x-1)^2$. [3]

(c) State the domain of the inverse you found in part (b). [1]

(d) Explain why $f(x) = x^3$ needs no restriction. [2]

24. Solve each equation.

(a) $|x-3| = 2$ [2]

(b) $|2x+1| = 7$ [2]

(c) $|3-x| = 1$ [2]

(d) $|x| = 5$ [1]

25. The graph of $y = f(x)$ crosses the x-axis at $x = 2$ and has a minimum point at $(0, -4)$.

(a) State where $y = |f(x)|$ meets the x-axis. [1]

(b) State where $y = [f(x)]^2$ meets the x-axis. [1]

(c) State the point that $(0, -4)$ becomes on $y = |f(x)|$. [2]

(d) Explain why $y = f(|x|)$ is always an even function. [2]

5.5 The Reciprocal Function HL

The last of these operations deserves a section of its own. Given any function f, its **reciprocal** is

$$y = \frac{1}{f(x)},$$

defined at every x where f is defined and $f(x) \neq 0$.

You never need to calculate points to sketch it. Every feature of the reciprocal can be read from the graph of f, one feature at a time.

Where f does this	The reciprocal does this	Reason
$f(x) = 0$	vertical asymptote	you cannot divide by zero
$\lvert f(x)\rvert$ is large	the curve approaches the x-axis	a large number has a small reciprocal
$f(x) = 1$ or -1	the two graphs cross	1 and -1 are their own reciprocals
$f(x) > 0$	$\frac{1}{f(x)} > 0$	the sign is preserved
a local maximum	a local minimum	and a minimum becomes a maximum

The last row is worth a moment. Where f is large the reciprocal is small, so a high point of f becomes a low point of $\frac{1}{f}$. This holds wherever f is not zero, whatever the sign of f.

CAUTION $\dfrac{1}{f(x)}$ *is not* $f^{-1}(x)$. *The reciprocal divides* 1 *by the output; the inverse undoes the function. For* $f(x) = x + 3$ *the reciprocal is* $\dfrac{1}{x+3}$ *and the inverse is* $x - 3$*, which are not the same. The notation* f^{-1} *never means "one over* f*".*

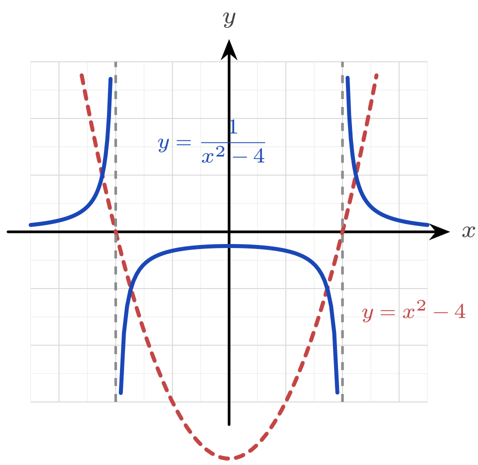

Figure 5.6 $y = \dfrac{1}{x^2-4}$ against $y = x^2 - 4$. Zeros of the original become vertical asymptotes at $x = \pm 2$, the minimum $(0, -4)$ becomes the maximum $(0, -\frac{1}{4})$ of the middle branch, and large values of f become values close to zero.

Check each rule against the figure. The dashed parabola is f and the solid curve is its reciprocal. The roots of f at $x = \pm 2$ have become vertical asymptotes. The minimum of f at $(0, -4)$ has become a maximum of $1/f$ at $(0, -\frac{1}{4})$. And where f takes large values, the reciprocal approaches the x-axis.

EXAMPLE 5.13

The graph of $y = f(x)$ has a single root at $x = 3$ and a local minimum at $(1, 2)$, and $f(x) \to$

∞ as $x \to \pm\infty$. Describe the corresponding features of $y = \dfrac{1}{f(x)}$.

Take the features one at a time.

The root at $x = 3$ makes the denominator zero, so $y = \dfrac{1}{f(x)}$ has a **vertical asymptote at** $x = 3$.

The minimum at $(1, 2)$ has $f(1) = 2$, so the reciprocal passes through $(1, \frac{1}{2})$. A minimum of f becomes a **maximum** of the reciprocal, so this is a local maximum at $(1, \frac{1}{2})$.

Since $f(x) \to \infty$ at both ends, $\dfrac{1}{f(x)} \to 0$, so $y = 0$ is a **horizontal asymptote**.

No algebra was needed. Every answer came from one feature of f.

TIP *Sketching a reciprocal in an examination: mark the roots of f as vertical asymptotes first, then the points where $f = 1$ and $f = -1$, then the turning points with their y-coordinates inverted. Draw the curve through those, one branch at a time.*

YOUR TURN 5.5 The Reciprocal Function HL

26. State the equations of the vertical asymptotes of $y = \dfrac{1}{f(x)}$ for each f.

(a) $f(x) = x - 4$ [1]

(b) $f(x) = x^2 - 9$ [2]

(c) $f(x) = (x - 1)(x + 5)$ [2]

(d) $f(x) = x^2 + 1$ [2]

27. Consider $y = \dfrac{1}{x^2 - 4}$.

(a) State the equations of the vertical asymptotes. [2]

(b) State the equation of the horizontal asymptote. [1]

(c) Find the value of y when $x = 0$. [1]

(d) State whether that point is a maximum or a minimum, and give a reason. [2]

28. The graph of $y = f(x)$ has a single root at $x = 3$ and a local minimum at $(1, 2)$, and $f(x) \to \infty$ as $x \to \pm\infty$.

(a) State the equation of the vertical asymptote of $y = \dfrac{1}{f(x)}$. [1]

(b) Find the coordinates of the turning point of $y = \dfrac{1}{f(x)}$, and state its type. [3]

(c) Describe the behaviour of $y = \dfrac{1}{f(x)}$ as $x \to \pm\infty$. [2]

(d) State the condition on $f(x)$ at a point where $y = f(x)$ and $y = \dfrac{1}{f(x)}$ cross. [2]

29. Sketch $y = \dfrac{1}{f(x)}$ for each function below, on separate axes. Show every asymptote and the coordinates of any turning point.

(a) $f(x) = x - 2$ [3] (b) $f(x) = x^2 - 1$ [4]
(c) $f(x) = x^2 + 1$ [4] (d) $f(x) = (x-1)(x-3)$ [4]

30. Let $f(x) = x - 2$.

(a) On one set of axes, sketch $y = f(x)$ and $y = \dfrac{1}{f(x)}$. [3]

(b) Find the coordinates of the two points where the graphs cross. *(Hint: they cross where $f(x) = \pm 1$)* [3]

(c) State the interval on which $\dfrac{1}{f(x)} > f(x)$ for $x > 2$. [3]

Chapter 5 · Function Operations

Reflections

TOK Slide the graph of $y = x^2$ two units right and you get $y = (x-2)^2$. Is this the same function or a different one? It has the same shape, the same name ("a parabola"), yet a different rule and different values everywhere. When we call two things "the same" in mathematics, we are always choosing what to ignore. The transformation makes that choice visible.

IM In 1872 Felix Klein, aged just twenty-three, proposed in his *Erlangen Programme* a radical reorganisation of geometry: a geometry *is* a set of transformations together with the properties they leave unchanged. Euclidean geometry studies what survives rotation and reflection; other geometries allow other moves. The transformations of this chapter are the same idea in miniature, and the question "what stays the same when I move this graph?" is one of the deepest in mathematics.

RW Transformations are the working language of digital media. Audio software shifts pitch (a horizontal stretch of a waveform) and adjusts volume (a vertical stretch); image editors scale, reflect, and translate pictures pixel by pixel; every animation moves one stored model through a stream of transformations. Engineers even split signals into an even part and an odd part, because each behaves differently in a circuit.

EXT Every function whatsoever splits, uniquely, into an even part and an odd part:

$$f(x) = \underbrace{\tfrac{1}{2}\big(f(x) + f(-x)\big)}_{\text{even}} + \underbrace{\tfrac{1}{2}\big(f(x) - f(-x)\big)}_{\text{odd}}.$$

Check it: the first bracket is unchanged when $x \to -x$, the second changes sign, and together they add back to $f(x)$. A single line of algebra, and every function is revealed as a sum of an even part and an odd part.

End-of-Chapter Problems — Chapter 5: Function Operations

1. Let $f(x) = 2x - 3$ and $g(x) = x^2$.

(a) Find $(f \circ g)(x)$. [2]
(b) Find $(g \circ f)(x)$. [2]
(c) Solve $(f \circ g)(x) = 5$. [2]

2. The graph of $y = f(x)$ has a maximum point at $(1, 4)$. State the coordinates of the maximum of:

(a) $y = f(x) + 2$; [1]
(b) $y = f(x - 3)$; [1]
(c) $y = 2f(x)$; [1]
(d) $y = -f(x)$ (state the resulting minimum). [1]

3. Let $f(x) = x^2 - 4x + 5$.

(a) Write $f(x)$ in the form $(x - h)^2 + k$. [2]
(b) Describe the two transformations mapping $y = x^2$ to $y = f(x)$. [2]
(c) State the range of f. [2]

4. Let $f(x) = \dfrac{x+1}{x-2}$, $x \neq 2$. Find $f^{-1}(x)$. [4]

5. Solve the equation $x^2 + 2 = 3x$, and hence solve the inequality $x^2 + 2 \leq 3x$. [5]

6. Determine whether each function is odd, even, or neither:

(a) $f(x) = x^3 - 2x$; [1]
(b) $f(x) = |x|$; [1]
(c) $f(x) = x^2 + x^3$. [1]

7. Consider $f(x) = x^2 + 2$ with domain $x \geq 0$.

(a) Find $f^{-1}(x)$. [3]
(b) State the domain of f^{-1}. [1]

8. Sketch the graph of $y = |x^2 - 1|$, clearly showing the x- and y-intercepts and the shape. [4]

9. Let $f(x) = x + 2$ and $g(x) = x^2$. Solve $(f \circ g)(x) = (g \circ f)(x)$. [4]

10. The graph of $y = f(x)$ passes through $(0, 1)$, $(2, 0)$ and $(3, 4)$. Find the images of these three points on $y = f(x + 1) - 2$. [3]

11. Let $f(x) = 3x + 2$.

(a) Find $f^{-1}(x)$. [2]
(b) Solve $f(x) = f^{-1}(x)$. [3]

12. Let $g(x) = \sqrt{x - 1}$.

(a) State the domain and range of g. [2]
(b) Find $g^{-1}(x)$ and state its domain. [3]

13. Let $f(x) = \dfrac{2}{x - 3}$.

(a) State the equations of the vertical and horizontal asymptotes. [2]
(b) Find $f^{-1}(x)$. [3]

14. The graph of $y = x^2$ is transformed to $y = -(x - 1)^2 + 2$. Describe the sequence of transformations and state the coordinates of the vertex. [4]

15. Solve the equation $|x - 2| = |x + 4|$. [4]

16. Solve the inequality $|2x - 1| < 5$. [3]

17. Determine whether each function is odd, even, or neither:

(a) $f(x) = \dfrac{x}{x^2 + 1}$; [1]
(b) $f(x) = \dfrac{1}{x^2 + 1}$; [1]
(c) $f(x) = x + 1$. [1]

18. The function f is even with $f(3) = 7$, and g is odd with $g(2) = -5$. Write down the values of $f(-3)$ and $g(-2)$. [2]

19. Let $f(x) = x^2 - 6x$.

(a) Write $f(x)$ in completed-square form. [2]
(b) Hence describe the transformations from $y = x^2$ and state the vertex. [3]

20. Find the coordinates of the point(s) where $y = |x|$ meets $y = 2 - x$. [3]

21. Let $f(x) = \dfrac{1}{x + 2}$.

(a) State the domain and range. [2]
(b) State the equations of the two asymptotes. [2]

22. The function $f(x) = ax + b$ satisfies $f(1) = 5$ and $f(3) = 11$.

(a) Find a and b. [3]
(b) Find $f^{-1}(x)$. [2]

23. The quadratic $f(x) = x^2 + bx + c$ has vertex $(2, -1)$.

(a) Find b and c. [3]
(b) Describe the transformation of $y = x^2$ that gives $y = f(x)$. [2]

24. Let $f(x) = x^2 - 2x$.

(a) Find the roots and the vertex of f. [3]
(b) Sketch $y = |f(x)|$, showing the intercepts and the turning point. [3]

25. Solve the inequality $\dfrac{2x-1}{x+2} \leq 1$. [5]

26. Let $f(x) = (x+1)^2 - 4$ for $x \geq -1$.

(a) Explain why f has an inverse on this domain. [1]
(b) Find $f^{-1}(x)$, stating its domain. [4]

27.

(a) Solve $|x - 2| = |3x + 1|$. [4]
(b) Hence, or otherwise, solve $|x - 2| \geq |3x + 1|$. [3]

28. For $f(x) = \dfrac{x}{1+x}$, find $(f \circ f)(x)$ and simplify fully. [5]

29. Prove that if f and g are both odd functions, then $f \circ g$ is also odd. [5]

30. The function $f(x) = ax + b$ satisfies $f(f(x)) = x$ for all real x. Find all possible values of a and b. [6]

31. Prove that every function $f : \mathbb{R} \to \mathbb{R}$ can be written as the sum of an even function and an odd function. [6]

32. Show that $f(x) = \dfrac{2x+3}{x-2}$ is self-inverse, that is, $(f \circ f)(x) = x$. [5]

33. Let $f(x) = x^2 - 4x + 3$.

(a) Write $f(x)$ in completed-square form, and state the coordinates of the vertex. [3]
(b) Find the roots of f and the y-intercept. [3]
(c) Sketch $y = f(x)$, showing the vertex and both intercepts. [2]
(d) On separate axes, sketch $y = |f(x)|$ and $y = f(|x|)$, showing the key points of each. [4]
(e) The graph of $y = f(x)$ is translated 2 units to the right and 3 units up to give $y = g(x)$. Find $g(x)$ in the form $ax^2 + bx + c$. [3]

34. Let $f(x) = \dfrac{3x-1}{x+2}$ for $x \neq -2$, and let $g(x) = x + 2$.

(a) State the range of f, justifying your answer. [2]
(b) Find $f^{-1}(x)$, and state its domain. [4]
(c) Find $(f \circ g)(x)$, giving your answer as a single fraction. [3]
(d) State the domain of $f \circ g$. [2]
(e) Solve $(f \circ g)(x) = 2$. [4]

35. HL Let $f(x) = x^2 - 4$.

(a) Sketch $y = f(x)$, showing the intercepts. [2]
(b) Write down the equations of the vertical asymptotes of $y = \dfrac{1}{f(x)}$, and explain how you found them. [3]
(c) Find the coordinates of the maximum point of $y = \dfrac{1}{f(x)}$ on the interval $-2 < x < 2$, and explain why it is a maximum even though the value is negative. [3]
(d) Describe the behaviour of $y = \dfrac{1}{f(x)}$ as $x \to \pm\infty$, and state the horizontal asymptote. [2]
(e) Sketch $y = \dfrac{1}{f(x)}$, showing all asymptotes and the point found in part (c). [4]

Challenge Questions

36. Self-inverse rational functions. A function of the form

$$f(x) = \frac{ax+b}{cx+d}, \qquad c \neq 0,$$

is called *self-inverse* when $(f \circ f)(x) = x$ for every x in its domain. This investigation finds exactly which of them are.

(a) Show that $f(x) = \dfrac{x+1}{x-1}$ is self-inverse. [3]
(b) Show that for the general function above,

$$(f \circ f)(x) = \frac{(a^2+bc)\,x + b(a+d)}{c(a+d)\,x + (bc+d^2)}.$$

[4]

(c) Hence prove that, when $c \neq 0$, the function f is self-inverse if and only if $a + d = 0$. *(Hint: two expressions in x are equal for all x only when their coefficients match)* [5]

(d) Show that if $ad - bc = 0$ then f is constant, and explain why the condition $ad - bc \neq 0$ must therefore be assumed throughout. [3]

(e) Write down two further self-inverse examples with $c \neq 0$, one of which has $b = 0$. Verify one of them. [3]

37. When does the order of two transformations matter? Section 5.2 showed that a vertical stretch and a vertical translation give different curves in different orders. This investigation settles the general question.

Throughout, start from $y = f(x)$.

(a) Apply a vertical stretch of factor a and then a translation of k units up, and write down the result. Do the same in the opposite order. Deduce the exact condition on a and k under which the two agree for every function f. [4]

(b) The horizontal moves are substitutions: move A replaces x by $x - h$, and move C replaces x by bx. Show that A followed by C gives $y = f(bx - h)$, while C followed by A gives $y = f(bx - bh)$. Deduce the condition under which these agree for every f. [5]

(c) Show that any vertical move and any horizontal move commute, whatever their parameters. *(Hint: consider which variable each move acts on)* [3]

(d) Using parts (a) to (c), state a single rule that predicts when two transformations may be applied in either order. Explain why your rule is consistent with all three cases above. [4]

38. Counting the solutions of $|f(x)| = k$. Let $f(x) = x^2 - 4$ and let $k \geq 0$ be a constant.

(a) Sketch $y = |f(x)|$, showing the x-intercepts and the coordinates of the local maximum. [3]

(b) Find the number of real solutions of $|f(x)| = k$ for each of $k = 0$, $k = 3$, $k = 4$ and $k = 6$. [4]

(c) Explain, using your sketch, why the number of solutions changes at $k = 4$ and not at any other value. [3]

(d) State the number of real solutions of $|f(x)| = k$ for every $k \geq 0$, giving your answer as a list of cases. [3]

(e) Now let g be any quadratic with a single minimum point of value m, where $m < 0$. State the corresponding list of cases for $|g(x)| = k$, in terms of m. [3]